Aircraft parameter estimation using a new filtering technique based upon a neural network and Gauss-Newton method

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ABSTRACT

A new parameter estimation method based upon neural network is proposed. The method proposed here uses feed forward neural networks to establish a neural model that could be used to predict subsequent time histories given the suitable measured initial conditions. The proposed neural model would not represent a generic flight dynamic model. The neural model in this case develops point to point fitting of the input and the output data. Thus, it could at best be referred to as flight dynamic model in restricted sense. Gauss-Newton method is then used to obtain optimal values of the aerodynamic parameters by minimising a suitable defined error cost function. The method has been validated using longitudinal and lateral-directional flight data of various test aircraft. The results thus obtained were compared with those obtained through wind tunnel test, or those obtained using Maximum likelihood and/or Filter error methods. Unlike, most of the parameter estimation methods, the proposed method does not require a prior description of the model. It also bypasses the requirement of solving equations of motion. This feature of the proposed method may have special significance in handling flight data of an unstable aircraft.

NOMENCLATURE

a_x, a_y, a_z	accelerations component along *x, y, z* body axes
b	span length
$\bar{c}$	mean aerodynamic cord length
C_l, C_m, C_n	rolling, pitching and yawing moment coefficient
C_D, C_L	drag and lift coefficient
C_X, C_Y, C_Z	force coefficient along *x, y, z* body axes
e^j	column vector with one in the jth row and zeros elsewhere
F_{eng}	thrust force by aircraft engine
I_x, I_y, I_z	moments of inertia about *x, y* and *z* axes
I_{xz}	moment of inertia about plane
m	mass of the aircraft
p, q, r	roll, pitch and yaw rates
$\dot{p}$, $\dot{q}$, $\dot{r}$	roll, pitch and yaw accelerations
$\bar{q}$	dynamic pressure
S	reference wing area
$\mathbf{U}(k)$	input vector at k^{th} instant
V	true airspeed
W	weight matrix in neural network
$Y(k)$	neural network out put vector at k^{th} instant
$\mathbf{Z}(k)$	measured output vector at k^{th} instant
α	angle-of-attack
β	angle of side-slip
ϕ, θ, ψ	roll, pitch and yaw angles
Θ	system parameters
δ_a, δ_e, δ_r, δ_f	aileron, elevator, rudder and landing-flap deflections
σ_{eng}	engine inclination angle with aircraft body axes
CG	centre of gravity
ENCG	distance between centre of gravity and engine

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1.0 INTRODUCTION

A new thrust area is emerging in the area of aircraft aerodynamic modeling and parameter estimation: development of techniques using artificial neural networks (ANNs) for the flight vehicle identification. More recently, many scientists and engineers have explored the potential of ANNs in diverse fields such as signal processing, pattern recognition, aircraft aerodynamic modeling, parameter estimation and control[1]. Artificial neural networks have been used in variety of applications because they are adaptive, they learn through examples and they can provide excellent functional approximation[1,2]. Recently, ANN modeling has been attempted for aircraft dynamics where aircraft motion variables and control inputs are mapped to predict the total aerodynamic coefficients[3-6]. In all these papers, the emphasis has been on aerodynamic modeling and estimation of aerodynamic coefficients using the Feed Forward Neural Networks (FFNNs). Raol and Jategaonkar[7] have used the Recurrent Neural Networks (RNNs) to model aircraft aerodynamics in a way that allows aircraft parameters to be estimated from flight data. However, as the authors[5] pointed out, the RNNs have only a limited scope of aircraft identification application and it is the FFNNs which may prove to be more flexible and thereby have a higher potential for future application for aircraft identification and parameter estimation.

Raisinghani, Ghosh *et al*[8] using FFNN proposed two new methods namely the Delta and the Zero method for explicitly estimating aircraft parameters from flight data. Both these methods do not require an *a priori* postulation of the model and bypass the requirement of solving equations of motion. The application of the Delta and the Zero method on the real flight data has been demonstrated in Refs 8-10. Further, the advantage of FFNN based methods in estimating parameters from flight data of unstable aircraft was also highlighted in Ref. 11. Application of the Delta and the Zero method to complete data yields N values, or in other words time histories, for the derivative (parameter) to be extracted. These extracted values are plotted as histograms, which usually show a near-normal distribution, from which the mean representing the aerodynamic derivatives can be determined[8,12]. Standard deviations are used as a measure of confidence in each parameter estimate.

A natural effort to improve the confidence level in the parameter estimate gets directed towards improving the FFNN training for modeling. There are several possible techniques including the more advanced Levenberg-Marquardt algorithm[13,14]. There exist many types of feed forward neural networks in the literature, for example, Multilayer Perceptron (MLP), Radial Basis Function (RBF) network[15,16] etc. Selectively these could be used to improve modeling using FFNN. However, due to lack of systematic principle/guideline to select the tuning parameters of FFNN, one may have to use his own judgment in freezing the tuning parameters. The application of various FFNN training algorithm generally shows a very fast decrease in cost function in the first few iterations, but the stringent tolerances for terminations are reached even after a few hundred iterations. This is attributed to the fact that data is corrupted with noise, and after capturing the main system characteristics, the network tries to account for random noise, leading to small oscillation in the cost function[12]. To avoid such phenomena, the Modified Delta method was proposed in Ref. 17, improving the Delta method (Modified Delta) following a different strategy in selecting input-output vector. The Modified Delta method was applied on real flight data pertaining to longitudinal and lateral-directional dynamics[17]. It was further shown that the Modified Delta method yielded parameter estimates with lesser standard deviation as compared to the estimates obtained through the Delta method. For all these methods standard deviations were used as a measure of confidence in the parameter estimates.

In the present work, a new method using FFNN is proposed for estimating the aircraft parameters from flight data. The proposed method uses FFNNs to establish a neural model that could be used to predict time histories of motion variables at $(k + 1)^{th}$ instant given the measured initial conditions corresponds to k^{th} instant. This method is christened the Neural Gauss Newton (NGN) method. The reasons for the nomenclature will become apparent once the method is described. The NGN method has been validated using real flight data pertaining to the longitudinal, and the lateral-directional dynamics of various test aircraft. Longitudinal flight data generated using test aircraft[18], Hansa-3 has been used for validation of the proposed method. Further the method has also been validated using lateral-directional flight data of ATTAS aircraft[12]. The results show that the NGN method has good potential as an alternative tool for parameter estimation from flight data.

The verification of the identified model is a key step in the identification process to assess the predictive capabilities of the extracted model. One approach is to compare the flight determined parameter estimates against the values obtained from the wind tunnel tests or analytical predictions. The other approach is known as the proof-of-match[12]. It is a widely used approach which is based upon the comparison of model predictions with flight measurements. To this end, flight data omitted from the identification studies is selected to ensure that the model is not tuned to specific data record or input form. In this study the verification of the identified model for longitudinal case has been carried out by comparing the flight extracted parameter estimates with the values obtained from wind tunnel tests[18]. Due to non-availability of the wind tunnel estimates of the lateral-directional parameters, the parameter estimates obtained from the flight data are compared for model verification with those reported in Ref. 12. A proof-of-match exercise has been carried out to verify the extracted model for both longitudinal and lateral-directional case. Finally, Filter Error Method[12,19] (FEM) has also been applied on these flight data to estimate the aerodynamic parameters. The effect of choice of the magnitude and the sign of the initial guess values of the parameters on estimated parameters have also been studied. Various runs were carried out with different sets of initial guess values of the parameter estimates. Runs were also carried out by changing both the signs and the magnitudes of the initial guess values of the parameters. It has been observed that the NGN method has the capability of extracting the aerodynamic parameters accurately even if the initial guess values are substantially off (both in magnitude and sign) from the nominal values of the aerodynamic parameters of the chosen aircraft.

2.0 NEURAL MODELING AND PARAMETER ESTIMATION

The estimation process starts with conducting of experiments. Predecided maneuvers are executed to excite selected dynamics of the chosen aircraft. On board data acquisition system is activated to acquire the flight data. These sets of flight data are referred to as measured flight data. The measured flight data undergo data compatibility check and all the motion variables are transferred to centrE of gravity (CG) of the chosen aircraft for further analysis. An algorithm using FFNNs and Gauss-Newton optimisation method has been proposed in this work to estimate aerodynamic parameters from flight data. The algorithm uses FFNNs to create neural model using time histories of motion and control variables of aircraft in flight. Once the neural model is validated, it can be used to compute response for any arbitrary control input. However, the proposed neural model does not represent a generic flight dynamic model. The proposed neural model can only be used to predict time histories of motion variables at $(k + 1)^{th}$ instant given the measured initial conditions corresponding to k^{th} instant (where $k = 1$ to n : n is the total number of discrete data point). It has been shown that for all practical purpose of parameter estimation, this approach helps in building flight dynamic model (in restricted sense) using measured input-output data and does not require any *priori* postulation of mathematical model or solving of equations of motion.

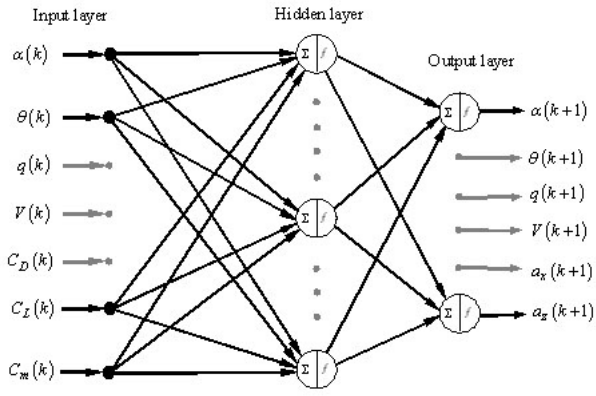


Figure 1. Neural-architecture for longitudinal flight dynamic model.

2.1 Neural Network model for longitudinal case

The scheme proposed to develop the neural model is illustrated in Fig. 1 for longitudinal case. Let the measured flight data contain the time histories at k^{th} instant of $\alpha(k)$, $\theta(k)$, $q(k)$, $V(k)$, $a_x(k)$, $a_z(k)$. Next step is to form the input $\mathbf{U}(k)$ and the output $\mathbf{Z}(k+1)$ vectors to be used for building the aircraft dynamic model using neural network architecture as given in Fig. 1.

The input vector $\mathbf{U}(k)$ and the output vector $\mathbf{Z}(k+1)$ required to build flight dynamic model are defined next.

$$\mathbf{U}(k) = \left[\alpha(k)\ \ \theta(k)\ \ q(k)\ \ V(k)\ \ C_D(k)\ \ C_L(k)\ \ C_m(k)\right]^T \quad \ldots(1)$$

Where the values of $C_D(k)$, $C_L(k)$ and $C_m(k)$ at the kth instant are obtained by substituting the relevant values of the flight variables into Equations (2 - 6).

$$C_D(k) = -C_X(k)\mathrm{Cos}\alpha(k) - C_Z(k)\mathrm{Sin}\alpha(k) \quad \ldots(2)$$

Where

$$C_X(k) = \left(ma_x^{CG}(k) - F_{eng}\mathrm{Cos}\sigma_{eng}\right)/\bar{q}(k)S \quad \ldots(3)$$

$$C_L(k) = C_X(k)\mathrm{Sin}\alpha(k) - C_Z(k)\mathrm{Cos}\alpha(k) \quad \ldots(4)$$

$$C_Z = \left(ma_z^{CG}(k) + F_{eng}\mathrm{Sin}\sigma_{eng}\right)/\bar{q}(k)S \quad \ldots(5)$$

$$C_m(k) = [I_y\dot{q}(k) - I_{xz}\left(p^2(k) - r^2(k)\right) - (I_z(k) - I_x(k))p(k)r(k) - F_{eng}\mathrm{Cos}\sigma_{eng}z_{ENCG} - F_{eng}\mathrm{Sin}\sigma_{eng}y_{ENCG}]/\left(\bar{q}(k)S\bar{c}\right) \quad \ldots(6)$$

The output vector $\mathbf{Z}(k+1)$ at $(k+1)^{th}$ instant required for the flight dynamic model is constructed in Equation (7).

$$Z(k+1) = \left[\alpha(k+1)\ \ \theta(k+1)\ \ q(k+1)\ \ V(k+1)\ \ a_x(k+1)\ \ a_z(k+1)\right]^T \quad \ldots(7)$$

Since the neural mapping uses measured motion variables, the performance and applicability of the proposed method can also be influenced by data quality. Special care needs to be taken in selecting tuning parameters, to avoid over training. Specially selection of number of iteration and number of neuron in hidden layer plays an important role during neural modeling while handling flight data with noise. Once the neural model is ready for prediction of motion variables, it is used to compute system output (Y) corresponding to assumed aerodynamic model (Θ) and measured initial conditions. Next, the difference between the measured response Z and system output Y is computed to estimate the noise covariance matrix R. Finally, the error cost function $J(\Theta)$ is minimised with respect to Θ by applying Gauss-Newton optimisation algorithm.

2.2 Algorithm of proposed NGN method

The algorithm of the proposed NGN method for parameter estimation is explained below with the help of block diagram in Fig. 2. The estimation procedure can be summarised as follows:

a) As a first step, the measured flight data undergoes data compatibly check. The measured motion variables are then transferred to center of gravity for further use during estimation process (Block; 1-3 of Fig. 2).

b) The procedure followed for neural training is explained in blocks 3-8. In this case FFNNs have been used for training.

c) Block 9 checks the convergence criteria for FFNNs training. Once training is accomplished, the trained neural model is used for parameter estimation.

d) Once aerodynamic model is chosen, the already trained neural model is used to calculate system output $Y(k)$. Input $\mathbf{U}(k)$ is constructed in block 10 using aerodynamic model fed through block 15. The input $\mathbf{U}(k)$ is fed to block 5 to estimate system output $Y(k)$.

e) The computed response $Y(k)$ and $\partial Y(k)/\partial\Theta$ (from block 11) are fed to block 12 and 13 to update aerodynamic parameter. The aerodynamic model is updated using this new set of aerodynamic parameters in block 15. The computation through steps (d) and (e) are continued till the convergence criteria (block 14) is met. Once convergence is achieved, parameters are estimated along with associated standard deviation.

The application of the proposed algorithm is explained next. For illustration, flight data pertaining to longitudinal case is considered. In the first step neural model is developed using input vector $\mathbf{U}(k)$ and output vector $\mathbf{Z}(k+1)$ for longitudinal parameter estimation. The input vector $\mathbf{U}(k)$ is reconstructed by keeping the same initial conditions ($\alpha(k)$, $\theta(k)$, $q(k)$, $V(k)$) used for training however, and corresponding to identical control input used for generating flight data are modified as per the chosen aerodynamic model in the estimating algorithm. Let, say, the chosen aerodynamic model for longitudinal parameter estimation be as given in Equations (8) to (10).

$$C_D = C_{D_0} + C_{D_\alpha}\alpha + C_{D_{\delta_e}}\delta_e \quad \ldots(8)$$

$$C_L = C_{L_0} + C_{L_\alpha}\alpha + C_{L_q}\left(q\bar{c}/2V\right) + C_{L_{\delta_e}}\delta_e \quad \ldots(9)$$

$$C_m = C_{m_0} + C_{m_\alpha}\alpha + C_{m_q}\left(q\bar{c}/2V\right) + C_{m_{\delta_e}}\delta_e \quad \ldots(10)$$

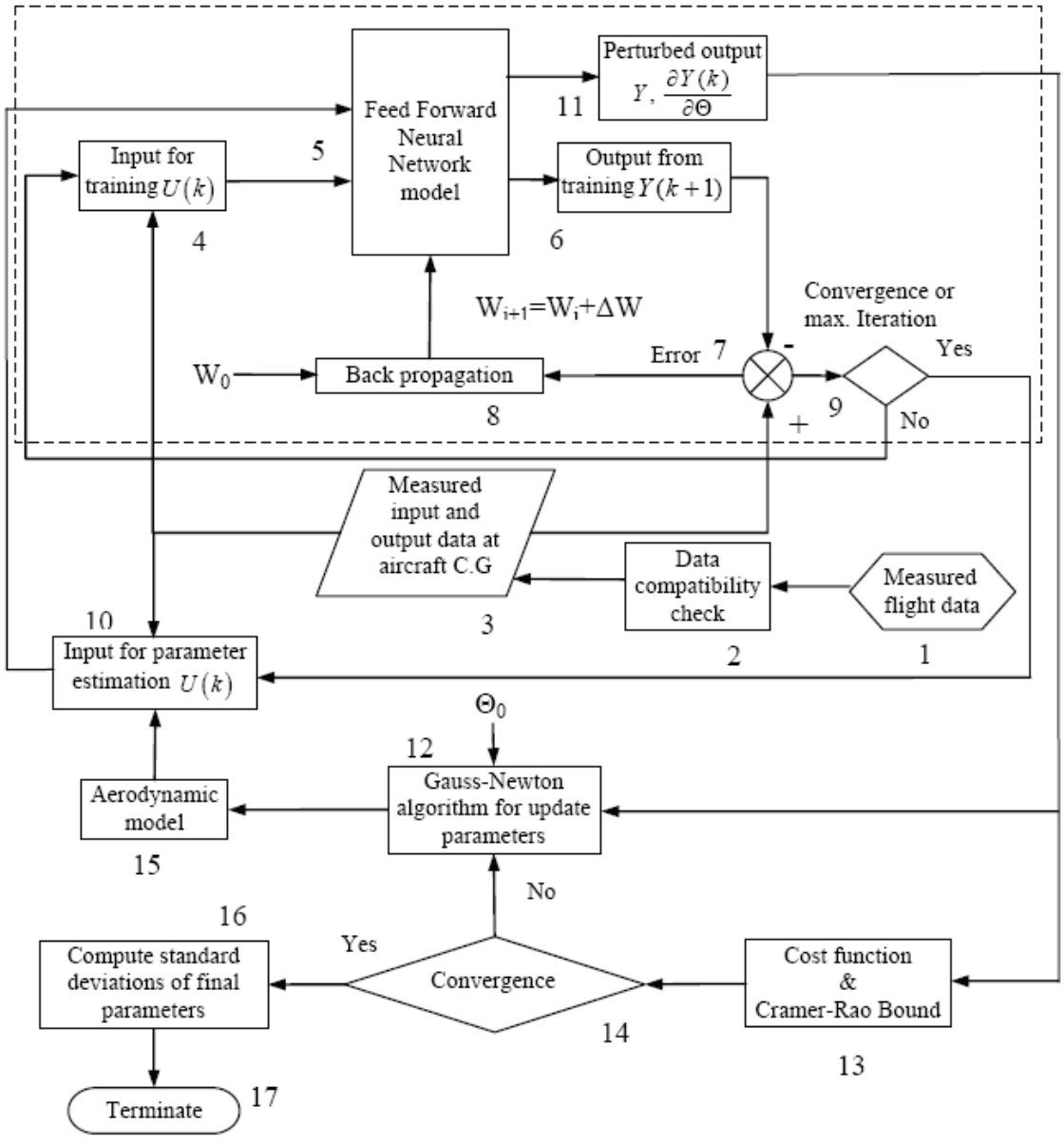


Figure 2. Schematic of NGN method.

The aim of the exercise is to estimate unknown parameter vector consisting of aerodynamic parameters namely, C_{D_0}, C_{D_α} ... C_{m_q} and $C_{m_{\delta_e}}$ as given in Equation (11).

$$\Theta = \left[C_{D_0} \; C_{D_\alpha} \; C_{D_{\delta_e}} \; C_{L_0} \; C_{L_\alpha} \; C_{L_q} \; C_{L_{\delta_e}} \; C_{m_0} \; C_{m_\alpha} \; C_{m_q} \; C_{m_{\delta_e}} \right]^T \quad \ldots(11)$$

To start the estimation algorithm, it is necessary to specify some suitable initial guess values of the unknown parameters vector Θ, consisting of non-dimensional parameters used for the description of aerodynamic model of C_D, C_L and C_m. Using the initial guess values of the aerodynamic parameters the estimated coefficients C_{De}, C_{Le} and C_{me} are computed using Equations (8) to (10). Next, the estimated response of the aircraft $Y(k + 1)$ at $(k + 1)^{th}$ instant is computed using input vector $\mathbf{U}(k)$ having same measured flight data (as given in Equation (7)) with C_D, C_L and C_m replaced by C_{De}, C_{Le} and C_{me}). Let this estimated response be represented by output vector as given in Equation (12).

$$Y(k+1) = \left[\alpha(k+1) \;\; \theta(k+1) \;\; q(k+1) \;\; V(k+1) \;\; a_x(k+1) \;\; a_z(k+1) \right]^T \quad \ldots(12)$$

The next task is to minimise the residual error $E(k)$ between the measure flight data $Z(k+1)$ and neural output $Y(k+1)$ and estimate the unknown parameter vector Θ. The residual error $E(k)$ and

covariance matrix of the residual R are computed using Equations (13) and (14) respectively. Next the values of $E(k)$ and R are used to compute error cost function $J(\Theta, R)$ using Equation (15).

$$E(k) = \left[Z(k) - Y(k) \right] \quad \ldots(13)$$

$$R = \frac{1}{N} \sum_{k=1}^{N} \left[Z(k) - Y(k) \right] \left[Z(k) - Y(k) \right]^T \quad \ldots(14)$$

$$J(\Theta, R) = \frac{1}{2} \sum_{k=1}^{N} \left[Z(k) - Y(k) \right]^T R^{-1} \left[Z(k) - Y(k) \right] \quad \ldots(15)$$

Finally, the Gauss-Newton method has been applied to update the parameter vector Θ by minimising the error cost function J(Θ, P) through the Equations (16) to (19).

$$\Theta_{i+1} = \Theta_i + \Delta\Theta, \quad \text{and} \quad \mathcal{F}\Delta\Theta = -\mathcal{G} \quad \ldots(16)$$

$$\mathcal{F} = \sum_{k=1}^{N} \left[\frac{\partial Y(k)}{\partial \Theta} \right]^T R^{-1} \left[\frac{\partial Y(k)}{\partial \Theta} \right] \quad \ldots(17)$$

$$\mathcal{G} = -\sum_{k=1}^{N} \left[\frac{\partial Y(k)}{\partial \Theta} \right]^T R^{-1} \left[Z(k) - Y(k) \right] \quad \ldots(18)$$

$$\left[\frac{\partial Y(k)}{\partial \Theta} \right]_{ij} = \frac{Y_{pi}(k) - Y_i(k)}{\partial \Theta_j} \quad \ldots(19)$$

The sensitivity matrix $[\partial Y(k)/\partial\Theta]_{ij}$ at each iteration (i) is computed using approximate relation given by Equation (19). The numerical values of Y_{pi} (perturbed response) are obtained by replacing parameter vector Θ with the $\Theta + \partial\Theta_j e^j$ (where e^j is column vector with one in the jth row and zeros elsewhere) in the input variable vector of the already trained neural model. The estimated error covariance matrix P is function of model parameters Θ, the data points being analysed and the covariance matrix of the residuals R. The expression to compute P is given in Equation (20)[12].

$$P \approx \left\{ \sum_{k=1}^{N} \left[\frac{\partial Y(k)}{\partial \Theta} \right]^T R^{-1} \left[\frac{\partial Y(k)}{\partial \Theta} \right] \right\}^{-1} \quad \ldots(20)$$

The standard deviations of the parameter estimates are the square root of diagonal elements of the estimation error covariance matrix P which can be computed using Equation (21).

$$\sigma_{\Theta i} = \sqrt{P_{ii}} \quad \ldots(21)$$

3.0 GENERATION OF FLIGHT DATA

3.1 Longitudinal flight data of Hansa-3 aircraft: FLT1 and FLT2

The final test of any scheme for parameter estimation must come from its successful demonstration on real flight data. A flight data base for identification studies was gathered from flight maneuvers with the test aircraft[18]. Typically, starting from trim flight conditions, the pilot applied control input in an attempt to excite the chosen dynamic modes. An onboard measurement system installed on the test aircraft provided measurements using dedicated sensors of a large number of signals such as aircraft motion variables, atmospheric conditions, control surface position etc. The measurements made in flight were recorded onboard using suitable interface with standard laptop. The flight data had raw data for measured V, α, β, p, q, r, a_x, a_y, a_z, ϕ, θ, Ψ, H, δ_e, δ_a, δ_r and the location of the measuring sensors. The measurement of airspeed (V), angle-of-attack (α), angle of side slip (β) were obtained with flight log mounted on a boom fixed to the tip of the wing. The angular rates p, q and r were obtained from the measurements available from the inertial platform. The accelerations along the three body axes were measured using an accelerometer triad located near the CG of the aircraft. The angular rates $\dot{p}$, $\dot{q}$ and $\dot{r}$ were obtained by numerical differentiation of the corresponding angular rates. The control surface deflections (δ_e, δ_a, δ_r) were measured using potentiometer. The temperature T was recorded using the standard cockpit outside air temperature (OAT) gauge. Two sets of flight data (FLT1 and FLT2) simulating short period longitudinal dynamics were generated at an altitude 6,000ft. The cruise speed at which the perturbations were initiated was fixed at nearly 56ms^{-1}.

The longitudinal flight data (FLT1) was generated using multi step elevator input ($\delta_{e_{max}}$ = 7deg) having total duration of 4s only. Another flight data set (FLT2) was generated with two similar looking double pulse input having almost same magnitude. These two pulses although looked similar but had opposite elevator deflections to excite the longitudinal dynamics. The proposed NGN method was applied on flight data FLT1 and FLT2 (Figs 3 and 4) to estimate the longitudinal aerodynamic parameters. Further, both these flight data were also used for model verification exercise using the proof-of-match technique[12].

3.2 Lateral-directional flight data of ATTAS aircraft: FLT3 and FLT4

The effectiveness of the NGN method for estimation of lateral-directional aerodynamic parameters was investigated using real flight data supplied by DLR Germany for the ATTAS aircraft[12]. The flight data for the flight condition defined by a landing-flap condition δ_f = special position (1 deg) and perturbation about steady state flight at indicated airspeed (IAS), V_{IAS} = 250kt were analysed. Unlike longitudinal parameter estimation, the lateral directional parameter estimation used two control inputs, namely aileron (δ_a) and rudder (δ_r). The selection of time interval between the application of δ_a and δ_r is extremely important[12,8]. Simultaneous application of δ_a and δ_r may result in poor estimation due to high correlation[12,9]. Further, only the application of will result in flight data rich in roll dynamics that could at best be suited for lateral parameter estimation. Similarly, dominant application of δ_r for perturbation will result in better estimation of directional parameters. A disjointed control input maneuver (with δ_a and δ_r not occurring simultaneously) would result in the better estimation of overall lateral-directional parameters. In view of the above, flight data FLT3 and FLT4 pertaining to lateral-directional dynamics were chosen for parameter estimation. The flight data FLT3 and FLT4 are presented in Figs 5 and 6.

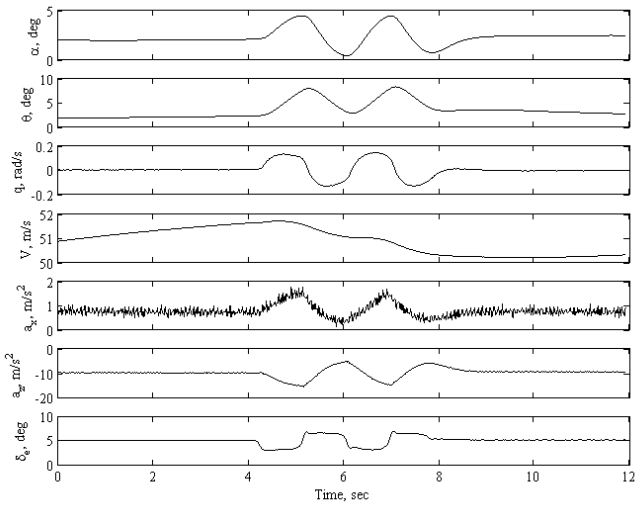


Figure 3. Longitudinal flight data: FLT1.

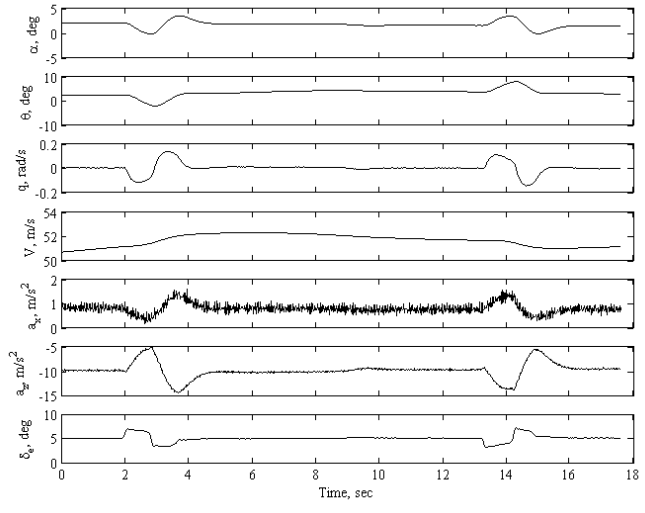


Figure 4. Longitudinal flight data: FLT2.

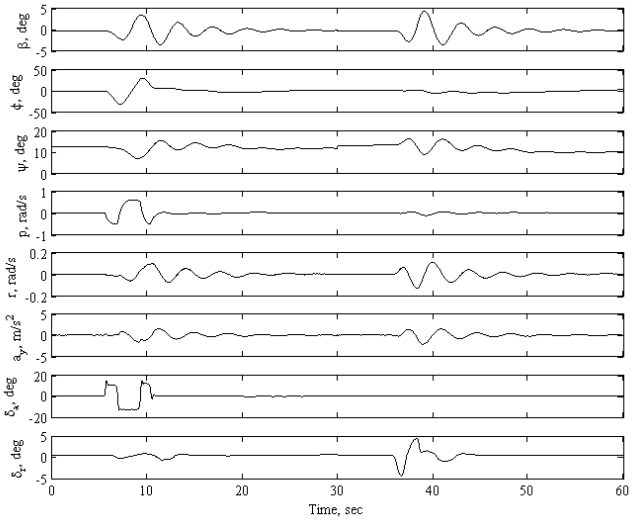


Figure 5. Lateral-directional flight data: FLT3.

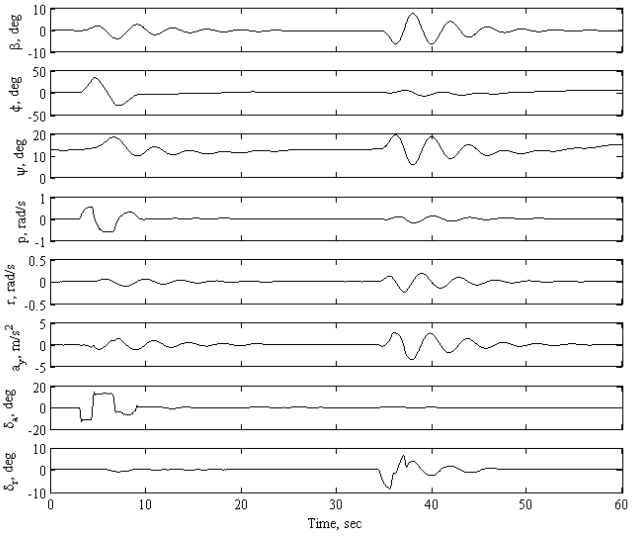


Figure 6. Lateral-directional flight data: FLT4.

4.0 APPLICATION OF THE PROPOSED METHOD TO REAL FLIGHT DATA

4.1 Longitudinal flight data FLT1 and FLT2 of Hansa-3 aircraft

The flight data FLT1 and FLT2 were generated by exciting the longitudinal dynamics of Hansa-3 test aircraft. The longitudinal short period dynamics was excited about a steady state trim at V = 56m/s at a cruise altitude of 6,000ft. Day and time of the experiment was carefully chosen to ensure fairly calm weather. The flight data acquired were preprocessed and reconstructed after carrying out exhaustive data compatibly check. Next, using these flight data, the input and output vectors for neural mapping were constructed. The input vector $\mathbf{U}(k)$ consisted of time histories at k^{th} instant of $\alpha(k)$ $\theta(k)$, $q(k)$, $V(k)$, $C_D(k)$, $C_L(k)$, and $C_m(k)$ whereas the output vector had $Z(k+1)$ and $\alpha(k+1)$, $\theta(k+1)$, $q(k+1)$, $V(k+1)$ and $a_z(k+1)$ as its elements. A typical FFNN structure used for building flight dynamic model has already been presented in Fig. 1. Using $U(k)$ as network input vector and $Z(k)$ as the output vector, the FFNN was trained as discussed earlier. The FFNN used a log-sigmoid and linear transfer function as the activation function and Levenberg-Marquardt[13,14] algorithm was used for updating the neural network weights. The mean square error (MSE) criterion or the number of iteration decided the termination of the iterative process. A range of values of the network parameters was tried to arrive at the final architecture of the FFNN used for parameter estimation. The network parameters varied were the number of hidden layers (1 – 3), the number of hidden neurons in the hidden layers (2–15), the learning rate (0·1 – 0·8), the momentum rate (0·1 – 0·8), and the number of iterations (100 – 2,000).

The network parameters finally chosen gave a good match between the true and the predicted values of the time histories of the variables. The final FFNN used had one hidden layer with ten neurons, the learning rate is 0·3 and number of iteration 300. The computation time is less than one minute.

The NGN method has then been used to estimate the stability and control derivatives (parameters) of the aircraft. Filter error method[12] was also applied on the flight data FLT1 and FLT2 to estimate longi-

Table 1
Hansa-3 Longitudinal Parameters

Parameter	Wind tunnel value	Estimated parameters from flight data			
		FLT1		FLT2	
		NGN	FEM	NGN	FEM
C_{D_0}	0·035	0·03645	0·036197	0·0286	0·03068
		(0·001284)*	(0·00182)*	(0·0015)	(0·00189)
C_{D_α}	0·0859	0·06091	0·070236	0·0374	0·00717
		(0·012539)	(0·01828)	(0·0149)	(0·01834)
$C_{D_{\delta_e}}$	0·0258	0·1515	0·15304	0·1867	0·16409
		(0·011986)	(0·01754)	(0·01477)	(0·01861)
C_{L_0}	0·354	0·229404	0·22091	0·2722	0·25843
		(0·004058)	(0·00627)	(0·00414)	(0·00505)
C_{L_α}	4·711	4·8856	4·7658	4·7145	4·6248
		(0·030049)	(0·05823)	(0·03199)	(0·04203)
C_{L_q}	-	37·25858	40·392	37·1543	42·1
		(1·1391)	(1·9943)	(1·3079)	(1·5673)
$C_{L_{\delta_e}}$	0·2653	0·37609	0·54365	0·31503	0·5979
		(0·050703)	(0·08072)	(0·0497)	(0·06029)
C_{m_0}	0·05214	0·0905124	0·083389	0·089796	0·08183
		(0·003072)	(0·00158)	(0·00304)	(0·00143)
C_{m_α}	-0·3372	-0·412282	-0·40089	-0·3777	-0·3672
		(0·026244)	(0·01461)	(0·02237)	(0·01153)
C_{m_q}	-	-8·79214	-7·1649	-9·4836	-8·0692
		(0·98729)	(0·51061)	(0·92061)	(0·4426)
$C_{m_{\delta_e}}$	-0·6941	-0·73488	-0·65859	-0·78167	-0·69625
		(0·039717)	(0·02062)	(0·03595)	(0·01708)

** Values in parentheses indicate sample standard deviation.*

tudinal aerodynamic parameters. The values of the estimated parameters namely C_{D0}, $C_{D\alpha}$, $C_{D\delta_e}$, C_{L0} $C_{L\alpha}$, $C_{L\delta_e}$, $C_{m\alpha}$, C_{mq} and $C_{m\delta_e}$ along with their Cramer-Rao bounds are presented in Table 1. Column 3 and 4 of Table 1 present the values of the estimated parameters obtained by applying NGN and FEM respectively on flight data FLT1. Similarly, column 5 and 6 present parameter estimates obtained by applying the NGN and the FEM on flight data FLT2. Column 2 lists the numerical values of the parameters obtained using wind tunnel tests[18]. It could be seen in Table 1 that the values of the parameters obtained using the NGN method are in fairly close agreement with the values obtained using wind tunnel tests[18] and FEM. It may also be seen that few aerodynamic parameters namely are not well estimated. Their values differ significantly, from the estimates obtained through wind tunnel tests. This may be due to lack of information content in the flight data (FLT1 and FLT2). It was observed that despite the best effort of the pilot, it was not possible to generate flight data pertaining[12] to 3-2-1-1 type elevator input excitation.

It may be mentioned here that while applying FEM, the estimated values showed significant dependence on the choice of initial guess values of the parameters. In contrast the values of the parameters estimated through the NGN method did not depend either on the magnitude or sign of the initial guess values of the parameters and choice of initial process noise matrix. This is expected as the propose NGN method bypasses the requirement of solving equations of motion. To check for the robustness of the NGN method with respect to choice of initial guess values of the parameters, the proposed algorithm was applied on simulated flight data of unstable aircraft to estimate the aerodynamic parameters. Although, not reported in this paper, the proposed NGN method could successfully estimate all the values of the parameters with correct signs and magnitudes.

Next, we carried out the model verification using proof-of-match procedure. Both the flight data FLT1 and FLT2 were alternatively used for parameter estimation and verification using proof-of-match[12] procedure. To start with, aerodynamic parameters were

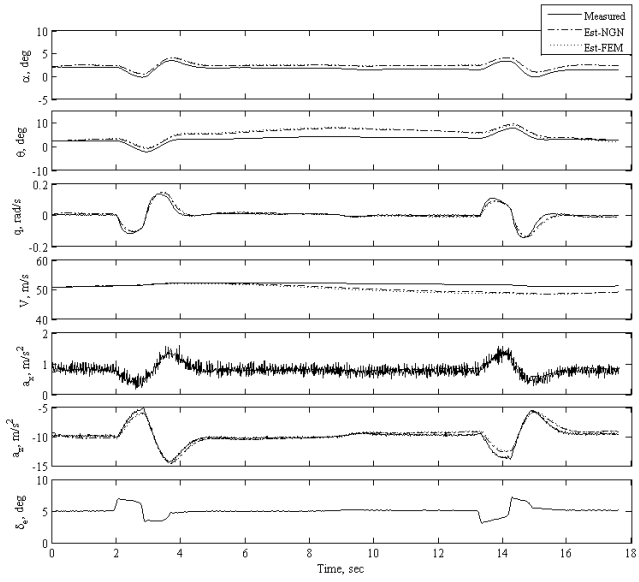


Figure 7. Model verification: parameter estimated using FLT1, proof-of-match using FLT2.

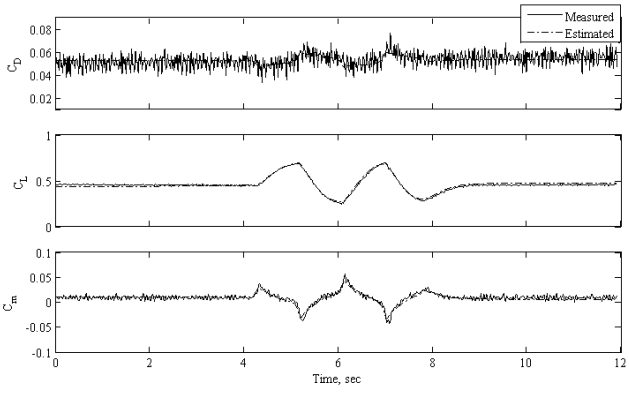


Figure 8. Comparison of aerodynamic coefficient: measured and estimated for FLT1.

estimated by applying the NGN and FEM methods on flight data FLT1. The estimated aerodynamic models were constructed by substituting the NGN and FEM estimates into Equations (8)-(10). These estimated aerodynamic model along with the initial conditions and control input used in generating real flight data FLT2 were used to compute estimated responses (Est-NGN, Est-FEM) by solving the equations of motion pertaining to longitudinal dynamics[12]. These estimated responses (Est-NGN and Est-FEM) were then compared with the measured flight data FLT2. A fairly close comparison among the measured flight data (FLT2) and estimated responses (Est-NGN and Est-FEM) has been observed as shown in Fig. 7. The estimated responses computed with FEM and NGN estimates are indistinguishable in the scale of Fig. 7. Similar exercise was also carried out wherein parameter were estimated using flight data FLT2 and for validation the estimated responses computed using the initial conditions and control input used in generating FLT1 were compared with the flight data FLT1. In this case also excellent matching was observed. Further, to see how good the estimated values are, we compared the estimated C_{De}, C_{Le} and C_{me} obtained by substituting the estimated values of the parameters obtained using NGN method in the right hand side of Equations (8)-(10), with the flight derived C_D, C_L and C_m being analysed. Figure 8 presents a

comparison between, C_L, C_D, C_m being analysed and estimated C_{Le}, C_{De}, C_{me}. The estimated model of C_{De}, C_{Le} and C_{me} matches closely with the flight derived values of C_D, C_L and C_m used in developing flight dynamic model using neural network.

4.2 Lateral-directional flight data FLT3 and FLT4 of ATTAS aircraft

The applicability of the NGN method was also investigated using real flight data pertaining to lateral-directional dynamics of the ATTAS aircraft. The flight data FLT3 and FLT4 as presented in Fig. 5 and Fig. 6 were used. The estimation of lateral-directional parameters in some sense was more involved as the flight data contained effects of two control inputs namely aileron, rudder and large number of parameters (as compared to longitudinal case) needed to be estimated.

The aerodynamic model pertaining to lateral-directional dynamics was assumed to have the following form.

$$C_Y = C_{Y_0} + C_{Y_\beta}\beta + C_{Y_p}\left(pb/2V\right) + C_{Y_r}\left(rb/2V\right) + C_{Y_{\delta r}}\delta r \qquad \ldots(22)$$

$$C_\ell = C_{\ell_0} + C_{\ell_\beta}\beta + C_{\ell_p}\left(pb/2V\right) + C_{\ell_r}\left(rb/2V\right) + C_{\ell_{\delta a}}\delta a + C_{\ell_{\delta r}}\delta r \qquad \ldots(23)$$

$$C_n = C_{n_0} + C_{n_\beta}\beta + C_{n_{\dot\beta}}\left(\dot\beta b/2V\right) + C_{n_p}\left(pb/2V\right) + C_{n_r}\left(rb/2V\right) + C_{n_{\delta r}}\delta r \qquad \ldots(24)$$

For lateral-directional case, the parameter vector Θ to be estimated is given in Equation (25).

$$\Theta = \left[C_{Y_0}\ C_{Y_\beta}\ C_{Y_p}\ C_{Y_r}\ C_{Y_{\delta_r}}\ C_{\ell_0}\ C_{\ell_\beta}\ C_{\ell_p}\ C_{\ell_r}\ C_{\ell_{\delta a}}\ C_{\ell_{\delta r}}\ C_{n_0}\ C_{n_\beta}\ C_{n_{\dot\beta}}\ C_{n_p}\ C_{n_r}\ C_{n_{\delta r}}\right]^T \qquad \ldots(25)$$

The input vector $U(k)$ and $Z(k+1)$ for lateral-directional case were constructed as given in Equations (26) and (27) respectively.

$$\mathbf{U}(k) = \left[C_Y(k)\ C_\ell(k)\ C_n(k)\ \beta(k)\ \phi(k)\ \psi(k)\ p(k)\ r(k)\right]^T \qquad \ldots(26)$$

$$\mathbf{Z}(k+1) = \left[\beta(k+1)\ \phi(k+1)\ \psi(k+1)\ p(k+1)\ r(k+1)\ a_y(k+1)\right]^T \qquad \ldots(27)$$

The values of $C_Y(k)$, $C_\ell(k)$ and $C_n(k)$ at the kth instant were obtained through the following equations.

$$C_Y(k) = ma_y^{CG} \big/ \bar{q}S \qquad \ldots(28)$$

$$C_\ell^{CG}(k) = \left[I_x\dot p - I_{xz}\dot r - I_{xz}pq - (I_y - I_z)qr - \Delta F_{eng}Sin\sigma_{eng}Y_{ENCG}\right] \big/ \bar{q}Sb \qquad \ldots(29)$$

$$C_n^{CG} = \left[I_z\dot r - I_{xz}\dot p + I_{xz}qr - (I_x - I_y)pq - \Delta F_{eng}Cos\sigma_{eng}y_{ENCG}\right] \big/ \bar{q}Sb \qquad \ldots(30)$$

Neural model representing the lateral-directional dynamics was arrived at using input vector $\mathbf{U}(k)$ and $\mathbf{Z}(k+1)$ as output vector. Once an acceptable neural model was established, the parameter estimation process was initiated with some initial guess value of parameter vector Θ. The parameters estimates along with their Cramer-Rao bounds obtained by applying NGN method on flight data FLT3 and FLT4 are presented in Table 2. It also lists the estimated values obtained using the maximum likelihood method[20], and these estimated values of lateral-directional parameters are treated as reference values of the parameters of ATTAS aircraft. Fairly good matching between the estimated values of the parameters with the reference values[20] validates the applicability of

Table 2
ATTAS lateral parameters

Parameter	Ref. 20	FLT3	FLT4
C_{Y_0}	-0·00348	-0·00869 (0·00012)*	-0·00908 (0·00009)
C_{Y_β}	-1·1562 (0·69)+	-1·1699 (0·00466)	-1·1569 (0·00306)
C_{Y_p}	0·2075 (8·12)	0·1396 (0·00827)	0·2605 (0·00756)
C_{Y_r}	0·5110 (6·96)	2·1034 (0·03616)	2·0399 (0·02174)
$C_{Y_{\delta r}}$	0·2225 (3·79)	0·2518 (0·0072)	0·2577 (0·00463)
C_{ℓ_0}	0·00069	0·0002 (0·00008)	0·000412 (0·00007)
C_{ℓ_β}	-0·1164 (0·91)	-0·1329 (0·003347)	-0·1385 (0·00233)
C_{ℓ_p}	-1·0139 (0·22)	-0·9423 (0·01005)	-0·9059 (0·01012)
C_{ℓ_r}	0·3195 (1·16)	0·3548 (0·02586)	0·36117 (0·01624)
$C_{\ell_{\delta a}}$	-0·2562 (0·20)	-0·2255 (0·001861)	-0·2215 (0·00199)
$C_{\ell_{\delta r}}$	0·0435 (2·16)	0·04575 (0·00546)	0·03983 (0·00357)
C_{n_0}	0·00135	0·00254 (0·00006)	0·00273 (0·00006)
C_{n_β}	0·2967 (0.11)	0·2566 (0.0019)	0·2759 (0.00208)
$C_{n_{\dot\beta}}$	-0·2839 (1·54)	-0·2011 (0·028782)	-0·287 (0·03409)
C_{n_p}	-0·0489 (1·20)	-0·0395 (0·003297)	-0·0506 (0·00458)
C_{n_r}	-0·5996 (0·94)	-0·5782 (0·03012)	-0·7504 (0·035001)
$C_{n_{\delta r}}$	-0·1691 (0·36)	-0·1314 (0·00321)	-0·1475 (0·003103)

+ Values in parentheses indicate standard deviation in %
* Values in parentheses indicate sample standard deviation.

the NGN method in estimating lateral-directional aerodynamic parameters from flight data. Few derivatives namely C_{Y_r}, C_{n_r}, estimated from FLT3 and FLT4 show scatter among themselves. However, interestingly not so strong derivatives, C_{n_β}, $C_{n_{\dot\beta}}$ are also well estimated along with lower Cramer-Rao bounds. Next, we carried out model verification using the proof of match procedure. In the first step, parameters were obtained using flight data FLT3. These parameters were fed in to the Equations (22)-(24) to arrive at the estimated model. The estimated model along with initial conditions pertaining to only aileron input of FLT4 were used to compute estimated response by solving equations of motion[12] pertaining to lateral-directional dynamics. Figure 9 shows comparison between the motion variables predicted using estimates obtained through NGN and MLE (Reference 20) methods. Similar exercise was also carried out where the estimates obtained by applying the NGN and the MLE methods on flight data FLT4, were used for computing estimated response corresponding to control input and initial conditions used in generating measured flight data FLT3. Figure 10 shows a comparison between the measured motion variables and estimated responses (Est-NGN, Est-MLE). For most of the cases the matching between the measured and estimated responses computed using the NGN and the MLE estimates was fairly good. Based on this study, it may be suggested that the proposed NGN method, which bypasses the requirement of solving the equations of motion, can advantageously be used as alternative tool for parameter estimation from flight data.

5.0 CONCLUSIONS

A new method christened the Neural-Gauss-Newton (NGN) method has been proposed for estimating aircraft parameters from flight data using feed forward neural networks. The proposed method advantageously uses the universal function mapping characteristics of feed forward neural network and optimization capability of Gauss-Newton algorithm. The results obtained for real flight data pertaining to longitudinal and lateral-directional dynamics have shown the success and the potential of the proposed method. Since the proposed method does not require solving the equations of motion, it presents itself as straight forward method in which the FFNN is trained to capture the flight dynamic model of an aircraft, and the parameters along with Cramer-Rao bounds are estimated in few iterations. The NGN method, which bypasses the requirement of solving the equation of motion, may have special significance in handling flight data of unstable airplane.

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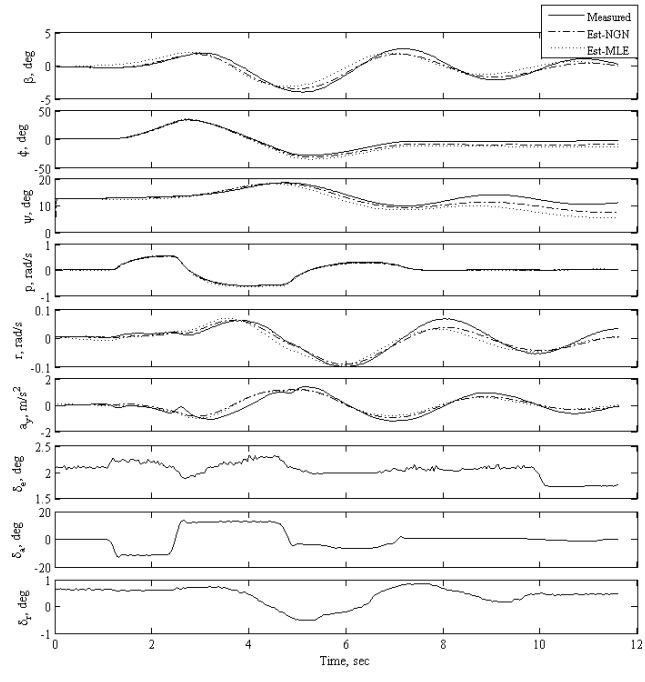


Figure 9. Model verification: Proof-of-match, aileron input of FLT4.

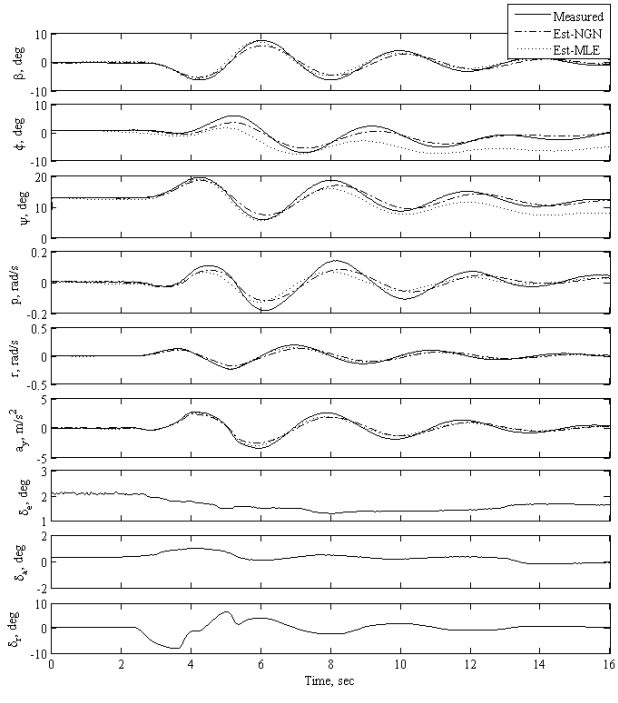


Figure 10. Model verification: Proof-of-match, rudder input of FLT4.

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